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3. Geometry of linear systems

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Given a system of 2 equations with 2 unknowns, each equation describes a line.

$$ax + by = 8$$

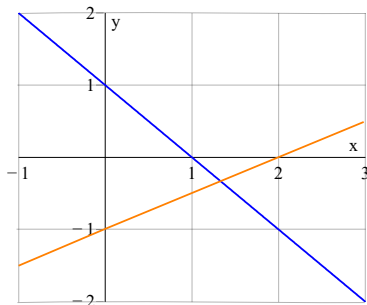
$$cx + dy = 2$$

The solution is the intersection of the solutions to each equation independently. There are 3 possibilities for the solution to such a system:

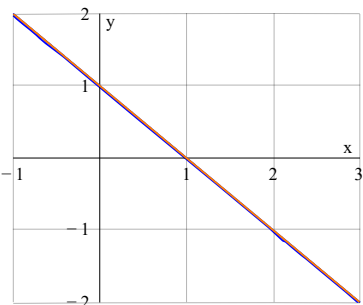
Example A: $x + y = 1$
 $x - 2y = 2$

Example B: $x + y = 1$
 $2x + 2y = 2$

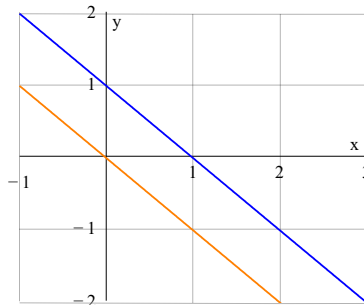
Example C: $x + y = 1$
 $x + y = 0$



In Example A (left), the two lines intersect at one point, this point is the solution.



In Example B (middle), the two lines are the same, so every point along the line is a solution.



In Example C (right), the two lines are parallel, never intersecting, and so there are no solutions to the system.

Given a system of 3 equations and 3 unknowns, it is more complicated – the solution can be a point, a line, a plane, or nothing!

Beyond equations with 3 unknowns, we have a hard time visualizing the solutions spaces geometrically. Instead, it is helpful to have algorithmic approaches to finding solutions. Part of this approach will involve writing our systems in terms of matrices. In this lecture, we will focus on developing an algorithm for finding solutions when there is one solution, a line of solutions, no solution. This same approach works in other situations, as seen in the complicated example at the end of this lecture.

3. Geometry of linear systems

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